

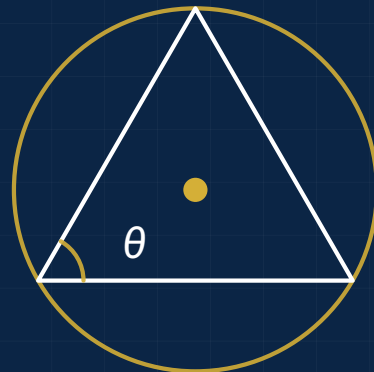


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CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 8

Congruence, Similarity & Proof

Similar shapes are enlargements: find the scale factor from one matching pair, then multiply or divide. Congruent shapes are identical — prove it with SSS, SAS, ASA or RHS.



3

VARIANTS

14

PAST-PAPER PARTS

29

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Similar shapes: scale factor

V2 Similar triangles in a diagram

V3 Proofs: congruence & similarity

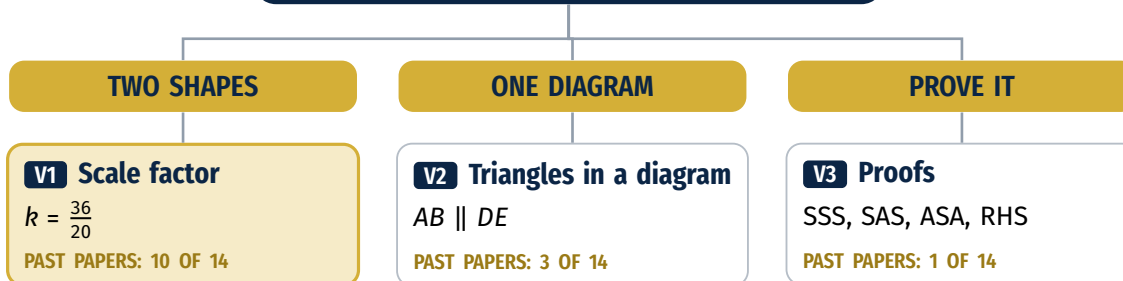
TOPIC MAP · EVERY VARIANT ON ONE PAGE

Similarity at a glance



Your notes

Congruence, Similarity & Proof



THE KEY HABIT

MATCH THE ANGLES FIRST

Corresponding sides face equal angles.

★ How often does each variant come up?

9 past-paper questions (Jan 2020 → Nov 2025) hold 14 parts; 29 practice parts — grouped by type. Area and volume of similar shapes are in Topic 9.

	PAST PAPERS (OUT OF 14)	PRACTICE PARTS
V1 Similar shapes: scale factor	10	12
V2 Similar triangles in a diagram	3	9
V3 Proofs: congruence, similarity	1	8

WHAT THIS TELLS YOU An early, quick question (Q3–Q12, 3–4 marks) on most papers: two similar shapes, find the missing sides. One scale factor, used carefully, is all it takes.

BEFORE YOU START

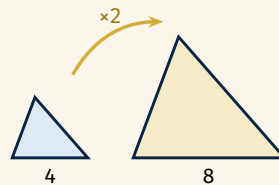
The big idea



Similar = same shape, different size.

All angles are equal, and every length is multiplied by the same **scale factor k** .

Congruent = same shape **and** size ($k = 1$).



Scale factor

Find k

$$k = \frac{\text{new}}{\text{old}}$$

From one matching pair.

Bigger

$$\times k$$

Small $\rightarrow$ large.

Smaller

$$\div k$$

Large $\rightarrow$ small.

Congruence conditions

SSS

3 sides

All three sides equal.

SAS

2 sides + $\angle$

The angle between them.

ASA

2 $\angle$ + side

A matching side.

RHS

90°

Right angle, hypotenuse, side.



CALCULATOR FIRST, THEN CHECK

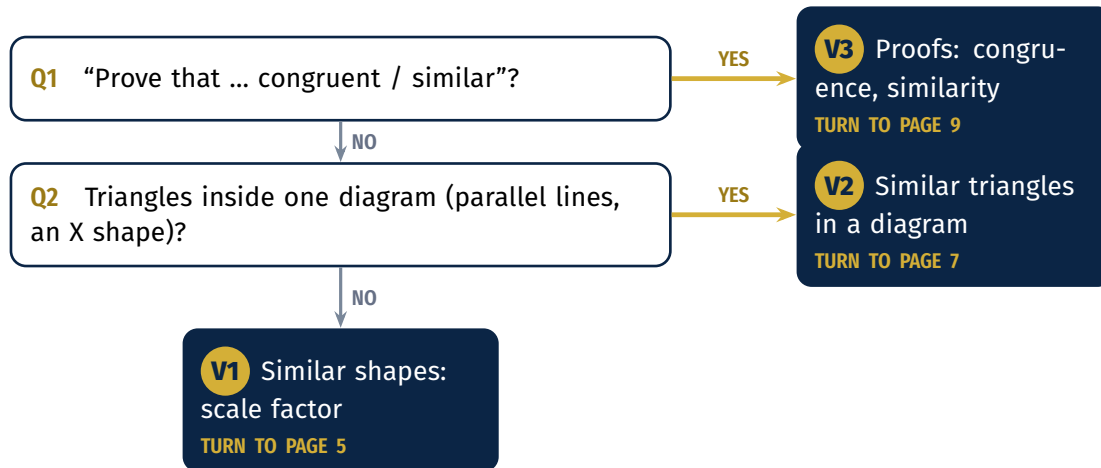
Check: the bigger shape's sides must all be bigger. If $k > 1$ gave a smaller side, you matched the wrong pair.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1 MOST COMMON IN THE EXAM

Similar Shapes: the Scale Factor

Match sides by equal angles; k from one known pair; multiply or divide.

SEEN IN PAST PAPERS

10 / 14

PRACTICE QUESTIONS: 12

 Your notes

Came up in: Nov 2021 P2H Q4 • Jan 2022 P2H Q3 • May 2023 P1HR Q3 • Nov 2024 P1H Q10 • Jun 2025 P2H Q5 • Nov 2025 P2H Q4

? What is it?

In plain words

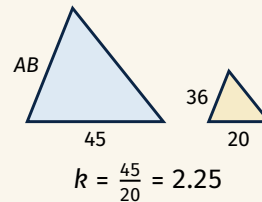
Find a pair of **corresponding** sides where you know both lengths — they face the same angle. $k = \frac{\text{big}}{\text{small}}$.

Small → big: $\times k$. Big → small: $\div k$.

IT LOOKS LIKE

$ABC \sim DEF$

similar quadrilaterals



How do I do it?

1 Match

Equal angles → corresponding sides.

2 Scale factor

$k = \frac{\text{big}}{\text{small}}$ from a known pair.

3 Use k

$\times k$ to the big shape, $\div k$ to the small.

4 Check

Big shape, bigger sides.

ONE PAIR GIVES k 10 of the 14 past-paper parts.

V1 Similar Shapes: the Scale Factor

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“ABC and DEF are similar triangles” → **Match by the angle marks**

“Write down an expression for y in terms of x ” → $y = kx$

Right-angled triangles with one side missing → **Pythagoras first, then k**

WORKED EXAMPLE **PAST PAPER**

November 2021 · Paper 2H · Q4 · 4 marks

ABC and DEF are similar. AC = 45 cm matches DF = 20 cm; DE = 36 cm.
(a) Find AB. (b) Given BC = 54 cm, find EF.

STEP 1 $k = \frac{45}{20} = 2.25$ ◀ scale factor

STEP 2 $AB = 36 \times 2.25 = 81$ ◀ (a) small → big

STEP 3 $EF = 54 \div 2.25$ ◀ (b) big → small

STEP 4 $= 24$ ◀

ANSWER (a) 81 cm (b) 24 cm

$$\checkmark \frac{81}{36} = \frac{54}{24} = 2.25 \checkmark$$

WATCH OUT

✗ $AB = 36 + 25 = 61$

✓ $AB = 36 \times 2.25$

Similar shapes multiply; they never add.

EXAM TIP

Write k as a fraction ($\frac{9}{4}$) to keep the arithmetic exact.

Check yourself

calculator first · answers on page 11

1.1 Jan 2022 P2H

ABC ~ PQR: AB = 4,
PQ = 12, RQ = 16.5,
AC = x , PR = y . Find BC
and y in terms of x .

Answer:

1.2 Nov 2024 P1H

Right-angled ABC
(AB = 7.5, AC = x) is
similar to DEF (DF = 24,
EF = 51, right angle at
D). Find x .

Answer:

1.3 Jun 2025 P2H

ABCD ~ EFGH: AB = 5,
BC = 4, CD = y ; EF = x ,
FG = 10, GH = 24. Find x
and y .

Answer:

V2

VARIANT 2

Similar Triangles in a Diagram

Parallel lines make equal angles, so the two triangles are similar; match the sides carefully.

SEEN IN PAST PAPERS

●●●●●●●●●● 3 / 14

PRACTICE QUESTIONS: 9



Your notes

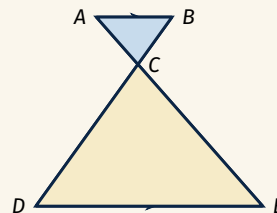
Came up in: Jun 2022 P2HR Q12 • Jun 2024 P2HR Q12

? What is it?

In plain words

Two shapes hide in one diagram: a small triangle **inside** a big one ($BE \parallel CD$), or an **X shape** ($AB \parallel DE$).

Parallel lines give equal angles, so the triangles are **similar**. Redraw them side by side, then use k .



IT LOOKS LIKE

 $AB \parallel DE$

ACE and BCD straight

How do I do it?

1 SpotParallel lines $\rightarrow$ equal angles (Z or F).**2 Redraw**

The two triangles separately, same way round.

3 Match

Which side faces which angle?

4 k or equationRatio of matching sides; solve if x appears.**REDRAW THEM APART** Then it is V1.

V2 Similar Triangles in a Diagram

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"ACE and BCD are straight lines; AB is parallel to DE" → **X shape: $ABC \sim EDC$**

"BE is parallel to CD" → **Small triangle inside the big one**

Lengths in x → **Ratio of sides = ratio of sides; solve**

WORKED EXAMPLE PAST PAPER

June 2022 · Paper 2HR · Q12 · 3 marks

ACE and BCD are straight; $AB \parallel DE$. $AC = 1.2$, $BC = 2x$, $CD = 2x + 9$, $CE = 6$. Find x .

STEP 1 $ABC \sim EDC$

◀ triangles: alternate angles, vertically opposite at C

STEP 2 $\frac{AC}{EC} = \frac{BC}{DC}$

◀ matching sides

STEP 3 $\frac{1.2}{6} = \frac{2x}{2x+9} \Rightarrow 0.4x + 1.8 = 2x$

◀ cross-multiply

STEP 4 $x = 1.125$

◀ solve

ANSWER $x = 1.125$ ✓ $\frac{2.25}{11.25} = 0.2 = \frac{1.2}{6}$ ✓**WATCH OUT**

$$\times \frac{1.2}{2x} = \frac{6}{2x+9}$$

$$\checkmark \frac{1.2}{6} = \frac{2x}{2x+9}$$

Each ratio must compare matching sides of the two triangles.

EXAM TIP

Label the matching vertices in the same order: $ABC \leftrightarrow EDC$.

Check yourself

calculator first · answers on page 11

2.1 Jun 2024 P2HR

$BE \parallel CD$, $AE = 10$,
 $CD = 1.5 \times BE$. Find ED ;
with $AB = 2x + 5$,
 $BC = 3x - 5$, find x .

Answer:

2.2 Dr Eslam

$DE \parallel BC$, D on AB , E on
 AC : $AD = 4$, $DB = 6$,
 $DE = 5$. Find BC .

Answer:

2.3 Dr Eslam

X shape: $AB \parallel DE$,
 $AC = 3$, $CE = 7.5$, $AB = 4$.
Find DE .

Answer:



VARIANT 3

Proofs: Congruence & Similarity

Three facts with reasons, then the condition (SSS, SAS, ASA, RHS — or AAA for similar).

SEEN IN PAST PAPERS

PRACTICE QUESTIONS: 8

1 / 14



Your notes

Came up in: Jan 2020 P2HR Q12

What is it?

In plain words

Congruent: show three matching facts (sides or angles) and name the condition.

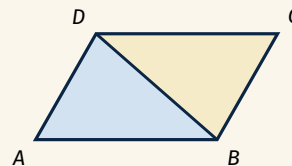
Similar: show the angles are equal (two pairs is enough — the third follows).

Every fact needs its **reason**: common side, alternate angles, radii, vertically opposite ...

IT LOOKS LIKE

ABCD parallelogram

chords crossing



SSS: $AB = CD, AD = CB, BD$ common

How do I do it?

1 Pick triangles

Name them in matching order.

2 Three facts

Sides / angles, each with a reason.

3 Condition

SSS, SAS, ASA, RHS (or AA for similar).

4 Conclude

"So the triangles are congruent (SSS)."

FACT + REASON, THREE TIMES Then name the condition.

V3 Proofs: Congruence & Similarity

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Prove that triangle ABD is congruent to triangle CDB ”

→ **Three facts + SSS/SAS/ASA/RHS**

“Show that the triangles are similar”

→ **Two pairs of equal angles**

WORKED EXAMPLE DR ESAM

Proof • 3 marks

$ABCD$ is a parallelogram. Prove that triangles ABD and CDB are congruent.

STEP 1+2 $AB = CD, AD = CB$

◀ opposite sides of a parallelogram

STEP 3 $BD = DB$

◀ common side

STEP 4 All three sides equal

◀ condition SSS

ANSWER triangles ABD and CDB are congruent (SSS) ✓ three facts, three reasons ✓

WATCH OUT

× “AAA” for congruence

✓ AAA only proves similar

Equal angles allow different sizes.

EXAM TIP

Write the vertices in matching order — it shows which sides you are pairing.

Check yourself

calculator first • answers on page 11

3.1

Jan 2020 P2HR

Isosceles triangles with base angles 66° ; at the apex the triangle and two congruent regular n -gons meet. Find n .

Answer:

3.2

Dr Esam

Two sides and the angle between them are equal. Which congruence condition?

Answer:

3.3

Dr Esam

One triangle has angles $40^\circ, 75^\circ$; another $40^\circ, 65^\circ$. Similar?

Answer:

CHECK YOURSELF

Answers



Your notes

Every answer below was checked with sympy (check_answers.py).

1.1 $BC = 5.5 \text{ cm}; y = 3x$

1.2 $x = 4$

1.3 $x = 12.5, y = 9.6$

2.1 $ED = 5 \text{ cm}; x = 3.75$

2.2 $BC = 12.5$

2.3 $DE = 10$

3.1 $n = 15$

3.2 SAS

3.3 Yes: angles
40, 65, 75

THE WHOLE TOPIC ON ONE PAGE

Similarity cheat sheet



Your notes
V1 Scale factor 10/14

$$k = \frac{\text{big}}{\text{small}}$$

× k up, ÷ k down.

V2 In a diagram 3/14

$$\frac{1.2}{6} = \frac{2x}{2x+9}$$

Parallel → similar; redraw apart.

V3 Proofs 1/14

SSS, SAS, ASA, RHS

Fact + reason; name the condition.

**TOP 3 LOST MARKS**

1. matching the wrong sides ×
2. adding instead of multiplying ×
3. AAA for congruence ×

BOTTOM LINE Match angles first · one pair gives k · parallel lines → similar · fact + reason.